Face Occlusion Prediction

Model Training Track — Data Challenge 2026

OccluVision (Group 8)

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https://github.com/LeHoang510/Face-Occlusion-Prediction

Introduction

Face occlusion estimation quantifies how much of a detected face is hidden (masks, hands, objects), which matters for downstream biometrics and fairness-aware systems. In the Model Training Track we regress a continuous occlusion ratio in $[0, 1]$ from 224×224 RGB face crops. The evaluation metric penalises errors on heavily occluded samples and rewards balanced performance across genders, so accuracy alone is insufficient. We pursued a three-phase pipeline: (i) benchmark diverse CNN/ViT backbones, (ii) mitigate gender disparity via balanced batching, and (iii) fine-tune a large self-supervised encoder with parameter-efficient adapters. Our submitted model—DINOv3-L + LoRA—ranks **3rd** on the interim leaderboard (score **0.00099**); code is available at the repository link above.

Task & Metric

The leaderboard score combines weighted accuracy and gender fairness (lower is better):

$$w_i = \tfrac{1}{30} + \mathrm{GT}_i, \quad \mathrm{Err} = \frac{\sum_i w_i (p_i - \mathrm{GT}_i)^2}{\sum_i w_i}, \quad \mathrm{Score} = \frac{\mathrm{Err}_F + \mathrm{Err}_M}{2} + |\mathrm{Err}_F - \mathrm{Err}_M|.$$

Weights penalise errors on heavily occluded faces; the disparity term encourages balanced performance across genders. All models optimise the same weighted MSE objective with standard augmentations (resize, flip, colour jitter). During development we hold out 10% of labelled data for validation (seed 42); the submitted checkpoint is retrained on the *full* training set.

Methods Explored

Phase 1 — CNN/ViT baselines. We fine-tune ImageNet-pretrained backbones from `timm/torchvision`: ResNet-50, ConvNeXt-Tiny, Swin-T, ViT-B/16, EfficientViT-B1, and DINOv2-Small. Each replaces the classification head with a dropout-regularised MLP ending in Sigmoid; all backbone weights are updated (up to 20 epochs, batch size 32, AdamW with cosine decay).

Phase 2 — Balanced-gender batching. To target the $|\mathrm{Err}_F - \mathrm{Err}_M|$ term, we use a custom sampler that builds each mini-batch with an equal number of male and female samples (minority class oversampled with replacement). We apply this to our strongest Phase-1 model (Swin-T) while keeping the same architecture and loss.

Phase 3 — DINOv3-L + LoRA. We adopt `facebook/dinov3-vitl16-pretrain-lvd1689m` (ViT-L/16), freeze the backbone, and inject LoRA adapters (r=16) on attention projections. Only LoRA weights and a regression head are trained—a parameter-efficient alternative to full fine-tuning. We train both a standard random sampler and a balanced-gender variant; Table 1 reports validation scores from the development split. The final submission retrains the best DINOv3-L + LoRA configuration on all labelled training data.

Submitted Solution: DINOv3-L + LoRA

Architecture. We freeze `facebook/dinov3-vitl16-pretrain-lvd1689m` (ViT-L/16, 1.1B params) and inject LoRA adapters (r=16, α=32, dropout 0.05) on `q_proj` and `v_proj`. The CLS token feeds a regression head: LayerNorm → MLP(512, GELU) → Linear → Sigmoid. Only ∼2% of parameters are trainable (LoRA + head).

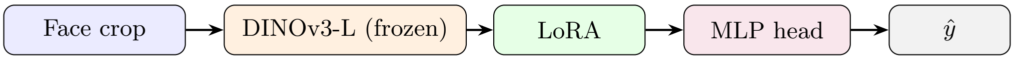

Figure 1: Submitted pipeline: frozen DINOv3-L backbone with LoRA adapters and a lightweight MLP head.

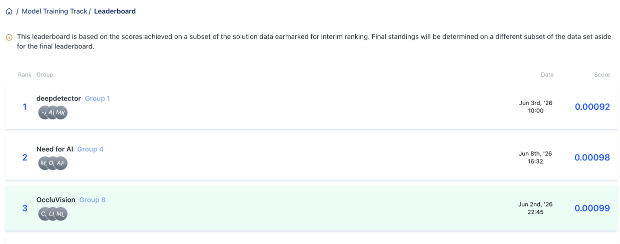

Figure 2: Interim leaderboard (final ranking uses a different test subset).

Training. We first tune hyper-parameters with a 90/10 train/val split, then retrain on the full labelled training set for submission. Training uses AdamW (lr=5×10^{-4}, weight decay 10^{-4}), cosine schedule with 1-epoch warmup, gradient clipping at 1.0; 15 epochs, batch size 64 on one RTX 3090 ($\sim$6 h). The development run reaches its best validation score at epoch 8 (Table 1); this configuration is reused for the full-data retraining that produces our leaderboard submission.

Results

Table 1: Best validation scores during model selection (90/10 split; lower is better).

Model	**Score**	Err_F	Err_M	**Ep.**
Phase 1 — CNN/ViT baselines				
Swin-T	0.001090	0.000481	0.000887	14
ConvNeXt-Tiny	0.001112	0.000671	0.000965	3
ViT-B/16	0.001162	0.000563	0.000962	8
ResNet-50	0.001221	0.000536	0.000992	11
EfficientViT-B1	0.001285	0.000660	0.001076	8
DINOv2-Small	0.001469	0.000693	0.001210	18
Phase 2 — balanced-gender batching				
Swin-T (balanced)	0.001150	0.000547	0.000949	10
Phase 3 — DINOv3 (submitted model in bold)				
DINOv3-L + LoRA	**0.001059**	**0.000567**	**0.000895**	8
DINOv3-L + LoRA (balanced)	0.001129	0.000525	0.000928	14

Limitations & Outlook

Limitations.

- **Compute:** Limited training budget—all experiments were capped at 20 epochs, which may leave performance on the table.
- **Hyper-parameters:** LoRA rank, learning rate, batch size, and epoch count all affect the fairness–accuracy trade-off; we did not run a full grid search.
- **Gender gap:** Despite the metric penalty, male error remains higher than female error on validation.

Future work.

- **Close the gender gap:** combine LoRA fine-tuning with balanced-gender batching (which lowered Err_F in isolation) and tune the loss to better match the $|\mathrm{Err}_F - \mathrm{Err}_M|$ penalty.
- **Ensemble:** average or stack predictions from DINOv3-L + LoRA with strong CNN/ViT baselines (e.g. Swin-T) to exploit complementary errors.